

5.2

$$M_2 = \{ (x, y, z)^T \in \mathbb{R}^3 : x - y + z = 0 \} =$$

$$= \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$M_3 = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} \quad A = [id]_B^K \mathcal{D} [id]_K^B$$

$$B = \left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right)$$

$$[id]_B^K = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$[id]_K^B = ([id]_B^K)^{-1} = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 & -1 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}$$

$$\det \begin{pmatrix} 3-\lambda & -1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & -1 & 3-\lambda \end{pmatrix} = (3-\lambda)^2(1-\lambda) + 2(3-\lambda) - (1-\lambda) - 2 =$$

$$= (9 - 6\lambda + \lambda^2)(1-\lambda) + 6 - 2\lambda - 1 + \lambda - 2 =$$

$$= 9 - 6\lambda + \lambda^2 - 9\lambda + 6\lambda^2 - \lambda^3 + 3 - \lambda =$$

$$= -\lambda^3 + 7\lambda^2 - 16\lambda + 12 =$$

z úkolu 5.1. $\Rightarrow$ 3 je vl. č

$$= -(\lambda - 3)(\lambda - 2)^2 = 0$$

$$A = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}$$